

Module Title:	6COSC017C - Machine Learning
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Feedback	

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Introduction

The chosen dataset shows where, at what time, and what kind of characteristics do people possess who committed major crimes such as homicide, shootings, robbery and serious assaults in Baltimore city. The data is updated every week and includes crimes from January 1, 2022 up to the present time. In 2025, Baltimore changed how it reports crime to follow a national system called NIBRS, which improves the accuracy and detail of crime reporting.

Link of the dataset (NIBRS Group A Crime Data):

<https://data.baltimorecity.gov/datasets/baltimore::nibrs-group-a-crime-data/about>

For this project, the intention of using this dataset is to predict what type of crime is likely to happen based on the spatial(*location*) and temporal(*time*) input features. Even though the original dataset has 21 input features, only 9 input features were manually selected while disregarding the rest of it. The reason being is that out of 21 features, 4 of them were related to personal characteristics (*such as age, gender, ethnicity and race*), and including them into the model would make the model biased towards people of particular features such as race. The other columns were simply a combination of the existing columns (*such as Geolocation was simply a concatenated column of Latitude and Longitude columns*), hence including them would simply be redundant whilst taking an unnecessary computational power. The other location related columns were disregarded due to them having a lot of incomplete entries, like Old and New Districts, and those values could anyways be inferred based on Latitude and Longitude.

The website for this project is hosted on the Streamlit Community Cloud, and the link along with the github repository is publicly available.

GitHub Repository: <https://github.com/00015775/MLDA-CW1-15775>

Streamlit App link: <https://mlda-cw1-15775-baltimore.streamlit.app/>

However, do note that if no people visit the website within certain days, approximately from 3 to 7 days without any visitors, then the link will switch to sleeping/hibernation mode. This does not mean that the website/link will stop working, but it does mean that it can take up from 30 seconds to a couple of minutes until the website is fully awake and is functional to use. Hence, give it some minutes for it to wake up.

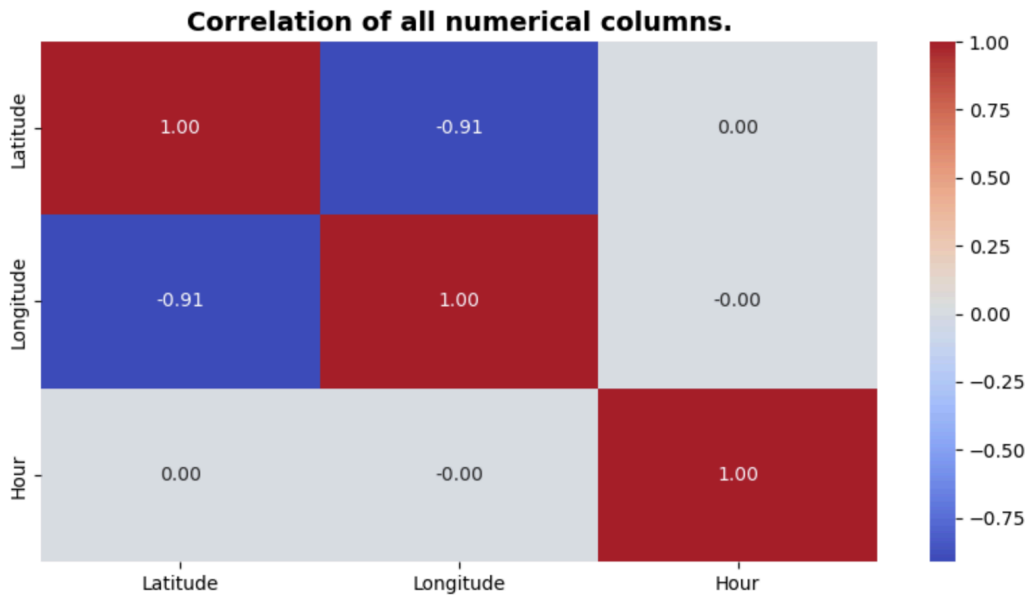
Description of Exploratory Data Analysis

The dataset has 228409 rows(*observations*) and 21 columns(*features*), but only 9 of those columns were selected as explained above. To be specific here is the list of those 9 columns along with their data types:

- **Latitude** - float64
- **Longitude** - float64
- **CrimeDateTime** - datetime
- **Month** - string
- **DayOfWeek** - string
- **Hour** - int32
- **PremiseType** - string
- **Inside_Outside** - string
- **Description**(*target label*) - string

When loading the dataset, the individual day, hour and month values were extracted from columns of CrimeDateTime due to their temporal relevance for this project. Since there are only 3 numerical input features, the correlation matrix was applied only to those columns, see the

matrix below:



In this case, Pearson correlation coefficient was applied which can only capture the linear relationship between two variables, so the values of 0 doesn't necessarily mean that there is no relationship at all, it could be hinting that there is a *non-linear relationship*, but in this project we stick to the linear relationship. Below is summary statistics of the numerical columns:

Summar statistics of numerical columns.

===== Latitude =====

Median: 39.30
Mean: 39.31
Std dev: 0.09
Variance: 0.01
Min: 0.00
Max: 39.38

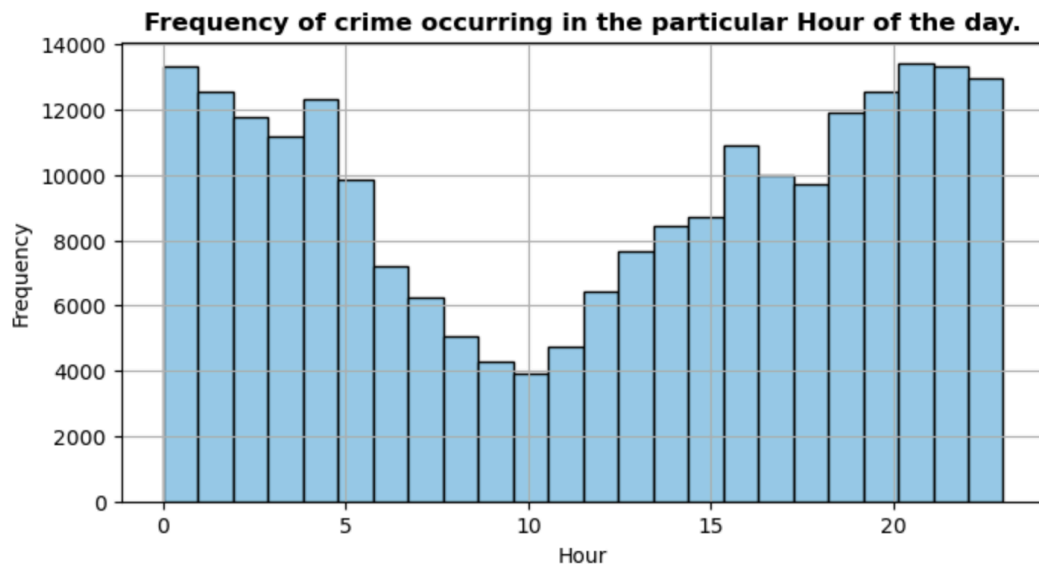
===== Longitude =====

Median: -76.61
Mean: -76.62
Std dev: 0.17
Variance: 0.03
Min: -76.73
Max: 0.00

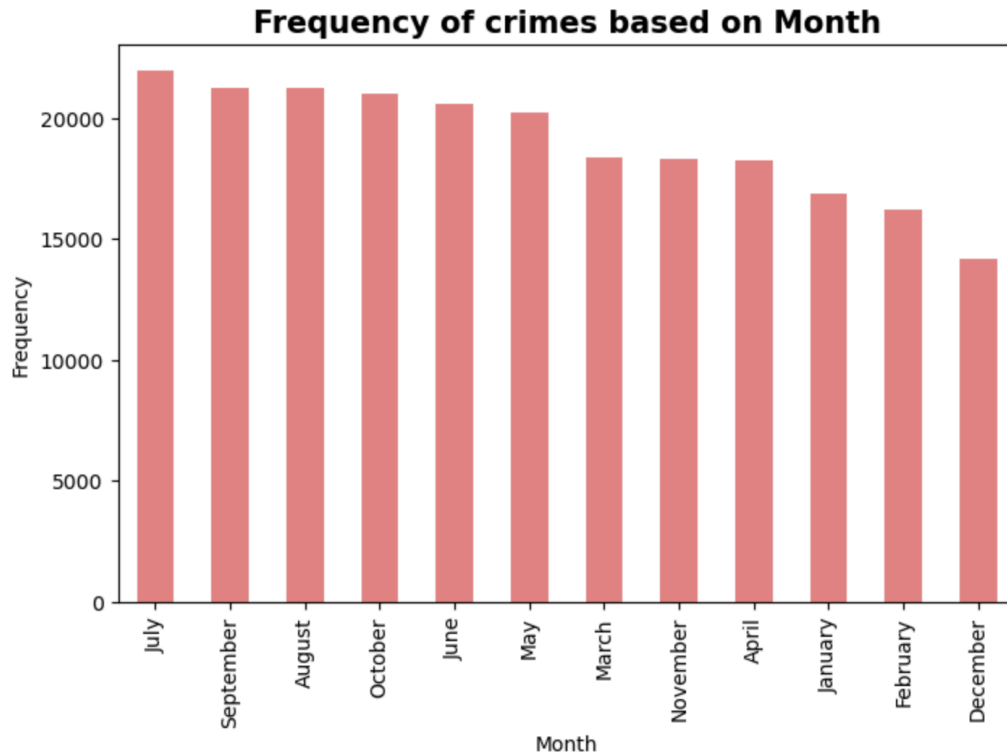
===== Hour =====

Median: 13.00
Mean: 11.91
Std dev: 7.72
Variance: 59.57
Min: 0.00
Max: 23.00

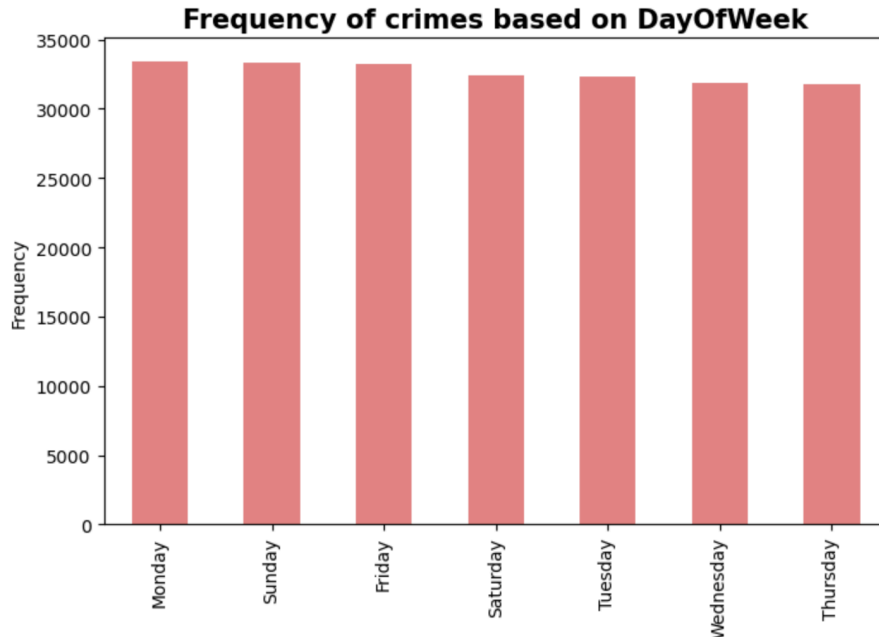
As the temporal information is of high importance for predicting the type of crime occurring at the particular time, the below given histogram represents the frequency of the crime occurring within the certain hours of the day:



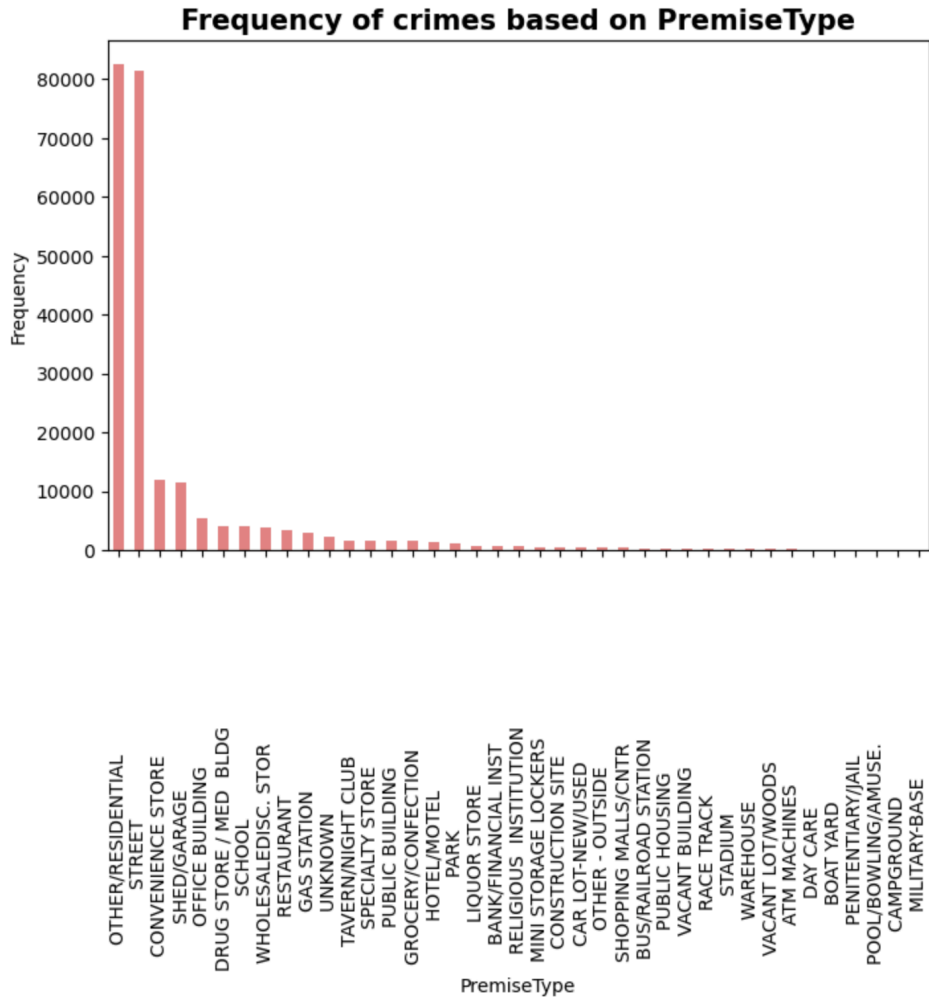
Here, we can see that most of the crimes start from the mid-nights between 00:00 and 05:00AM, but this tendency instantly plummets between sunrise and midday and slowly gets back to its peak close to night time from 18:00-23:00. In the following diagram, we can see that the month of the year does provide insights of the likelihood of the crime. As we can see, mostly during the summer and autumn seasons the most of the crimes happen, while during the winter and spring the frequency of crimes decrease by around 15-20%.



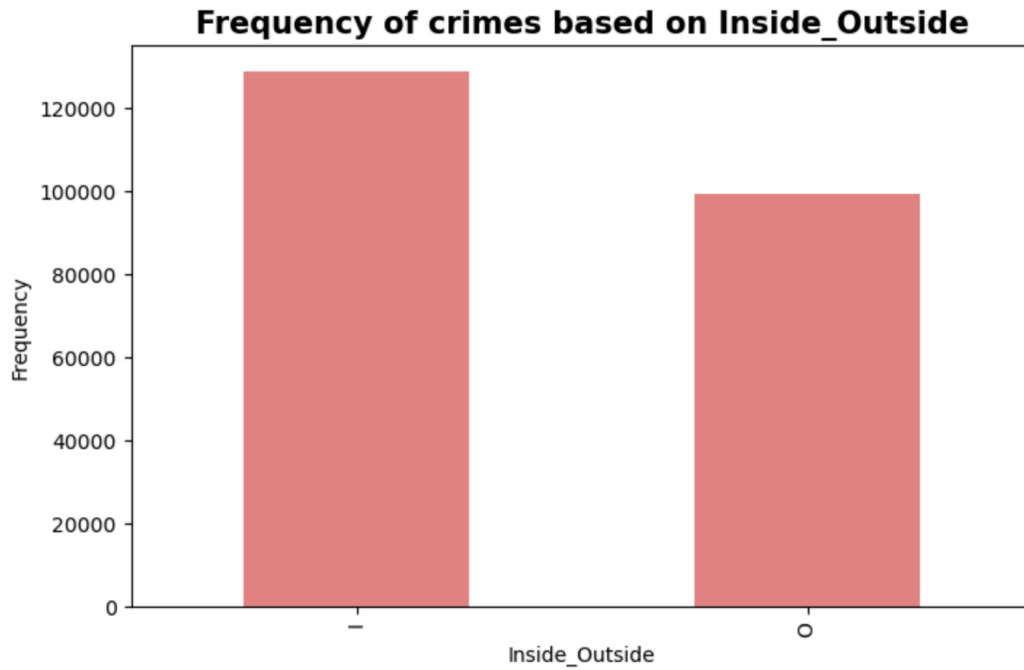
The next diagram shows the relationship of days of the week to the crime rates. Based on the diagram, while there is a tiny difference between the frequency of crimes between the days of the week, we still can see that crime rates for Monday, Sunday and Friday (*which are usually starting and ending schedules at the most workplaces*) are slightly higher compared to weekends. This information can provide us insights such as crime related to burglary or stealing from houses tend to occur when the owner of the house is at work.



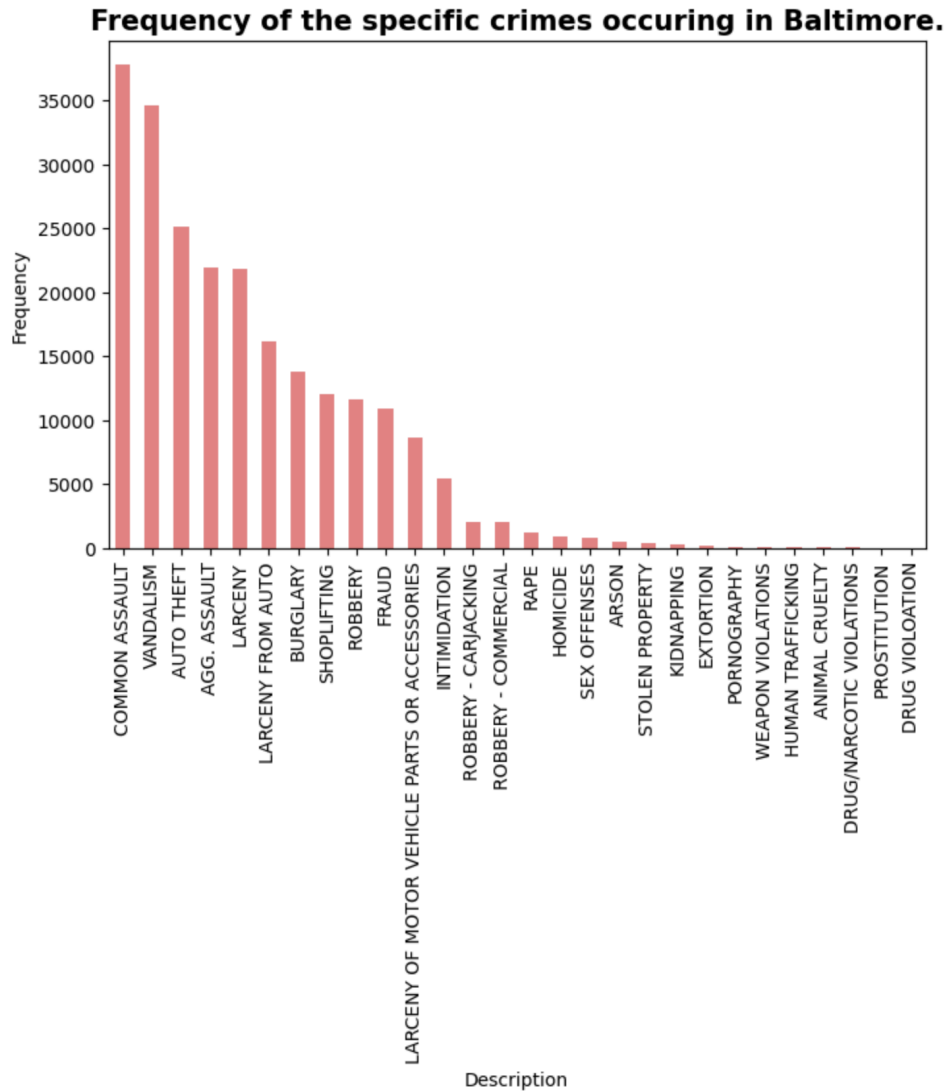
The above given three diagrams represent the importance of temporal relationship to the type of crime occurring in that specific timeframe. As for the spatial information, the next diagram shows the importance of how the location plays a role in predicting the crime. We can see that most of the crimes happen in the residential area, street, convenience and drug store, garage, school and even restaurants. While the rest of the locations do not exhibit as much crime rates as the above given premises, this diagram tells us that, given that we know the type of the premise, with some degree of confidence the model will be able to predict the type of crime occurring in that particular location.



While we can see that most crimes happen in the street, the following diagram shows us that the crimes happening indoors, such as inside of buildings, happen more often than from the outside of the premises. Around 20.000 more crimes happen inside of the buildings as opposed to being outside, which could seem counterintuitive.



The following diagram is about the frequency of the particular type of crime occurring, which is to say which crimes happen the most in Baltimore city and this value is what we want to predict in this project based on the discussed above input features.



Dataset Preparation

The rows with the missing values were completely dropped, and namely there were only 4 columns with missing or incomplete rows, which are Latitude, Longitude, Inside_Outside and PremiseType:

Latitude	262
Longitude	262
CrimeDateTime	0
Month	0
DayOfWeek	0
Hour	0
PremiseType	1
Inside_Outside	233
Description	0

The reason for dropping them are as follows:

- Latitude and Longitude are critical for spatial analysis and replacing them with mean/median would be incorrect
- Inside_Outside is related to location, so dropping it together.
- PremiseType has only one missing value, thus dropping its missing row will not affect model accuracy

In total 495 rows were dropped out of 228k rows. Since models expect numerical inputs, and only 3 columns were of integer type, the one-hot-encoding was applied to the rest of the 5 input features (*5, and not 6, because excluding the target label*). While the integer based encoding could have been applied to day and month, doing so would make the model assume that, for instance Tuesday is of higher importance than Monday and the same goes for the months and since this behaviour is not intended in this project, the ultimate choice was one-hot-encoding. The same encoding was applied to other input features as there was no inherent ordinal hierarchy in the names of the classes. Since PremiseType had 38 unique classes, it would take a lot of memory and computational power to train the dataset with it, furthermore after encoding, its dimensions would increase even more. Due to these concerns, only the top 10 most

frequent classes of the PremiseType column were chosen while putting the rest into the class of “OTHER”. Due to the dataset being very imbalanced, the outliers were completely dropped and out of 228K rows, around 200K were used for training and testing due to the hardware limitations of the computer. When the dataset was trained with outliers included, the overall performance was 20% at best for all three trained models, hence this result further justified the decision of dropping those rare crimes.

As mentioned above, the dataset is very imbalanced and in order to handle this, the upsampling and downsampling techniques were applied so that the number of classes within each row were more or less equal, without being biased towards the dominant values. After many iterations of training the model and evaluating its performance, it was found that 200K provided better results compared to training on 50K and 100K rows.

Justification of the choice of Machine Learning algorithms

When it came to choosing the model the choice leaned towards the one which was robust in accepting the target variable as of categorical data type. Initially LightGBM and XGBoost were experimented, but after finding out that the target categorical labels had to be converted into numerical ones(*which was not the intention*), after some experimentations the ultimate choice were Random Forest, Hist Gradient Boosting and Cat Boosting algorithms. Obviously, supervised learning was chosen for its higher level of control that it offers compared to unsupervised learning. For the hyperparameter tuning, the parameters of “*n_estimators*” were tuned with *GridSearchCV* to find the best values to define the total number of decision trees.

While the *GridSearchCV* was initially also applied to Gradient and Cat Boosting algorithms, at the later stages the choice leaned towards manually experimenting by finding the best values for the hyperparameters. This was done because initially *GridSearchCV* took up a lot of time to calculate the best hyperparameters for those algorithms and yet yielded poor results afterall.

As this project's nature is classification problem, the Recall, Accuracy, Precision and F-1 score were used as an evaluation metrics across all three trained models. The following is the final results of the model predicted outputs:

==== Random Forest ====

Accuracy: 0.6660

Precision: 0.6389

Recall: 0.6684

F1-score: 0.6514

==== HistGradientBoosting ====

Accuracy: 0.6543

Precision: 0.6201

Recall: 0.6567

F1-score: 0.6322

==== CatBoost ====

Accuracy: 0.6381

Precision: 0.5947

Recall: 0.6405

F1-score: 0.6109

Based on these evaluation results, Random Forest provided much better accuracy compared to other models trained in this project. Hence by further handling the imbalance nature of the dataset and applying different kinds of feature extraction and data cleaning, potentially higher accuracies can be achieved with Random Forest. As for the choice of the model in the deployed streamlit application, the Hist Gradient Boosting model was chosen due to it being much lighter(*70MB*) compared to Random Forest(*340MB*) and Cat Boosting(*96MB*).

Conclusion

In this project, it was found that the location and time does play an important role in identifying the type of crime happening. While the accuracy is not as high as expected, which can be explained by the high imbalance nature of the dataset alongwith the hardware limitations of the computer to train the model, overall it does provide a reasonable level of accuracy with at most 66% confidence of correctly identifying the type of the crime based on spatial and temporal data. The limitation of this project is that it boldly assumes that there is a linear relationship between the variables, while completely disregarding the possibility of existence of non-linear dependency. The future works can be tailored towards exploring and training with non-linear algorithms, which could potentially yield a much better performance and accuracy compared to linear models. Also, further data cleaning and handling imbalance distribution of classes can be applied to input features in order to yield higher accuracy. As for the ethical considerations, the gender, age, ethnicity and race were completely disregarded in identifying the crime as they would introduce bias towards people of certain characteristics.